MEGHA JOSHI

MSc Data Science Student | Python, SQL, Machine Learning, Dashboards
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PROFILE

MSc Data Science student at the University of Glasgow with a computer science degree. I work mainly in SQL and Python. My projects include an NHS waiting times dashboard built from five years of England and Scotland data, and an anti-money-laundering alert system that cut false positives by 98% on synthetic data. Looking for a graduate role in data analytics, data engineering or data science.

EDUCATION

Master of Science (MSc) in Data Science, University of Glasgow, UK Sep 2025 - 2027 (expected)
Modules: Information Visualisation, Statistical Methods, Machine Learning, Big Data Systems (Hadoop, Spark/PySpark), Text and Information Retrieval, Deep Learning, Large Language Model Principles

B.E. Computer Science and Engineering, Chandigarh University, India Aug 2020 - Jun 2024
Coursework: Database Systems, Data Structures and Algorithms, Object-Oriented Programming, Software Engineering

TECHNICAL SKILLS

Data Analysis: SQL (joins, subqueries, window functions), Python (pandas, NumPy, SciPy), exploratory data analysis, data cleaning and blending from multiple sources, MS Excel (pivot tables, charts)
Machine Learning: scikit-learn, LightGBM, PyTorch, SHAP; logistic regression, gradient boosting, Isolation Forest, deep learning, cross-validation, out-of-time validation, calibration
Visualisation and Reporting: Interactive dashboards (Plotly Dash, Streamlit), Tableau and Power BI (currently learning), matplotlib, data visualisation, explaining results to non-technical users
Databases and Tools: SQLite, MySQL, PySpark, Hadoop (coursework), Git, GitHub Actions, pytest, Jupyter, Microsoft Azure (AZ-900)

WORK EXPERIENCE

Junior Embedded Engineer, Radius Synergies International Pvt. Ltd., Noida, India Dec 2024 - Jun 2025
- Tested and validated utility meters and source code with a colleague before deployment, and wrote the technical documentation and test reports used in QA review.
- Wrote firmware in Embedded C for pre-metering systems on STM32G070, STM32F101CB, LPC812 and LM3S815 microcontrollers.

PROJECTS

NHS Waiting Times Dashboard | Python, SQL, Plotly Dash github.com/meghajoshi-ds/nhs-waiting-times
- Blended over five years of NHS England A&E and referral-to-treatment data (237 trusts, 141,504 RTT rows) with NHS Scotland data, then cleaned and standardised both sources into one SQLite database for analysis in SQL and pandas.
- Built an interactive dashboard for a non-technical audience: a patient picks a specialty and region, then sees the fastest providers and how many weeks they could save by switching.
- Used a Mann-Kendall trend test with Sen's slope and seasonal decomposition to show that 4-hour A&E performance has declined significantly ($p < 0.001$), and that trusts stronger on emergency care also do better on elective care ($r = 0.37$, 124 trusts).

AML Transaction Monitoring | Python, SQL github.com/meghajoshi-ds/AML-Transaction-Monitoring
- On 60,059 synthetic transactions, combined four detection rules with an Isolation Forest. False positives fell from 5,245 to 89 with all 139 planted laundering cases still caught. Alert precision rose from 2.6% to 61%, which saves about 1,290 analyst hours a year.
- Rewrote the rules in SQL with window functions, with tests that check SQL and Python flag exactly the same transactions. The 67-test suite runs in CI on pandas 2 and 3.

Credit Risk Scorecard | Python, LightGBM, SHAP, Streamlit github.com/meghajoshi-ds/credit-risk-scorecard
- Built logistic regression and LightGBM models to predict loan default on 163,987 Lending Club loans (18.3% default rate). LightGBM reached AUC 0.690 and KS 0.283, about one point ahead of logistic regression.
- Found that interest rate leaks the answer, as the lender sets it from its own risk model. Leaving it out cost 2.4 AUC points.
- Validated out-of-time on 396,030 dated loans (trained on 2007-2013, tested on 2015). Ranking held up (AUC 0.696), but predicted default rates came out 6-7 points too low, so flagged that a lender would need to recalibrate.
- Turned the model into a 7-characteristic WoE points scorecard and a Streamlit app that scores an applicant and explains the main reasons behind the score.

ACHIEVEMENTS, CERTIFICATIONS AND PUBLICATIONS

- CodeOlympics 2026 Gold Medal: 1st place as part of Team 29 (1,884.6 points), Glasgow University Tech Society
- Kaggle Bronze Medal: Kaggriculture competition, ranked 1,007th of 10,246 teams (top 10%)
- Certifications: Microsoft Certified: Azure Fundamentals (AZ-900); Software Testing (NPTEL); Fundamentals of Java (Infosys Springboard)
- Publication (peer-reviewed): Tokenizing Art: Unpacking the Impact of Non-Fungible Tokens on the Ownership Paradigm, IEEE Xplore, 2024